



Five Architectural Tests in the ESPR

The EU's Digital Product Passport regime binds five requirements on every certification verification system referenced in a DPP. Most existing systems clear one or two. Compliance requires all five.

As the Commission's July 19, 2026 registry deadline approaches, [EU Regulation 2024/1781](#) binds five architectural requirements on every certification verification system referenced in a DPP, set in [Article 9](#) with related provisions in Articles 10 and 11: open and non-proprietary, machine-readable, real-time, independently verifiable, and persistent. Two govern the format of the data. Three govern the behavior of the verification path. Under the [EU Battery Regulation](#), batteries face the first mandatory passports in February 2027. Each requirement binds independently, so the question for compliance teams is: *does the architecture clear all five?*

A verification system can clear two of these and fail three. Compliance requires all five.

OPEN AND NON-PROPRIETARY

Built on publicly available, internationally recognized standards: the Commission has signaled W3C decentralized identifiers, verifiable credentials, GS1 identifiers, and JSON-LD as the aligned primitives. Non-proprietary is a replicability test: a third party with no prior business relationship must be able to replicate verification access using publicly documented standards alone, without dependence on any commercial technology provider. The requirement reaches beneath the user interface, to the layers where the data flows. *A platform that requires a vendor SDK, a paid API tier, or a private contract fails the test.*

MACHINE-READABLE

The requirement most often confused with "online." The ESPR specifies data that is structured and searchable: consumable by software without human translation. Practical formats that satisfy it: [JSON](#), [JSON-LD](#), [XML](#), and W3C verifiable-credentials payloads. The test is whether a regulator's compliance software, an auditor's database tool, or a third-party service can retrieve and parse certificate status programmatically. *A rendered page that says "Status: Valid" satisfies neither requirement.*

The regulatory boundary lives at the architecture, not the user interface.

The three behavior tests

Real-time means the verification path reflects the certificate's current status at the moment of the check, per [Article 10](#): revocation, expiration, suspension, and scope change must propagate before a downstream user acts on a stale "valid." Most certification portals update on batch cycles measured in days or weeks; the ESPR moves the floor. Independently verifiable means a third party with no relationship to the certification body can verify through public means alone. The practical test is the disappearance check: if the issuer's staff were unavailable, could a regulator still verify? Persistent (Article 11) means the record remains accessible for the product's typical lifetime plus up to ten years, even through insolvency. The verification record must outlive any single commercial entity, including the certification body itself.

The Five-Question Architectural Test

A compliance team can clear the bar with five questions. 01: Can a third party with no prior relationship retrieve verification status programmatically, using only publicly documented standards? 02: Is the data machine-readable and structured: JSON, JSON-LD, XML, or W3C verifiable credentials? 03: Does the path reflect current status in real time, including revocation, expiration, and scope change? **The last two probe survivability.** 04: Can verification complete without any contact, confirmation, or assistance from the certification body or its platform operator? 05: Will the path remain accessible for the product's lifetime plus the regulatory minimum, even if the issuer or operator ceases operations? Yes to all five meets the standard. Any no fails it.

What compliance looks like

The architectural primitives that satisfy the standard are public and documented: [W3C verifiable credentials](#), [decentralized identifiers](#), [JSON-LD structured data](#), and [well-known URIs](#) for discovery. Real-time status propagates through revocation registries and timestamped records. Persistence comes from neutral, independently hosted infrastructure that does not depend on any single commercial entity's continued existence.

The simplest implementation path is neutral verification infrastructure that sits above existing certification platforms, exposing the required architecture without forcing a system migration. IntegraLayer was built for that role: a neutral verification layer that satisfies all five requirements while leaving the certification body's authority, branding, and program intact.

For certification bodies navigating the DPP category

IntegraLayer provides neutral verification infrastructure that sits above existing certification platforms. Independently verifiable in real time, on open standards, with your authority intact. Patent pending.

[Click here to get in touch →](#)

Trust isn't claimed. It's proven. The truth is verifiable.

IntegraLayer LLC | Patent Pending

Rachel Buchanan, Founder & CEO

[LinkedIn](#) | integralayer.io